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PROJECT SYNOPSIS

HEALTH HIVE

A Health-Focused Community Platform

Project Type: Mini Project / Learning Project

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1. ABSTRACT

Health Hive is a community-driven healthcare discussion platform designed to facilitate secure interaction between patients, healthcare enthusiasts, and verified medical professionals. The system integrates real-time communication, community engagement mechanisms, and role-based moderation to promote the dissemination of reliable health information while maintaining user safety and content integrity.

2. OBJECTIVES

- To build a full-stack health community platform that enables users to connect, share, and discuss health-related topics.
- To provide a secure and real-time communication system for discussions and direct messaging.
- To ensure content reliability through doctor verification and moderation mechanisms.
- To deliver a seamless and responsive user experience across devices.

3. PROPOSED SYSTEM

Health Hive consists of a scalable web application where users can join communities, create or participate in discussions, communicate via direct messages, and interact through real-time notifications. The platform includes an admin panel for content moderation and a doctor verification workflow to identify and badge verified medical professionals. The system architecture follows a client-server model with real-time capabilities using WebSockets.

4. KEY FEATURES

- User authentication with email verification & password reset
- Community discussions with nested replies
- Direct messaging (one-to-one)
- Real-time notifications
- Bookmark posts for quick access
- Doctor verification workflow
- Reporting and content moderation
- Admin panel for user and content management
- Responsive and user-friendly interface

5. TECHNOLOGY STACK

Frontend	React 18, Vite, Tailwind CSS, shadcn/ui, React Router, Socket.io Client, Recharts, Axios
Backend	Node.js, Express.js
Database	MongoDB (Mongoose)
Real-time Communication	Socket.io (WebSockets)
Authentication	JWT (JSON Web Tokens), bcryptjs
Other Services	Cloudinary (Image Uploads), Nodemailer (Email Services)

6. CURRENT DEVELOPMENT STATUS

The project is currently under active development.

7. FUTURE SCOPE

- Integration of AI-based health insights and recommendations
- Mobile application for enhanced accessibility
- Appointment scheduling with verified doctors
- Health resource library and multimedia support

Developer

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